

# Intelligent Capacity Planning for EMX

## Executive assessment

The proposed capability should be built as a **forecast-to-workforce decision engine**, not as another demand forecast, dashboard or route optimiser. Its job is to translate each forecast vintage into four operational decisions:

1. how many productive courier-hours are required at each store and interval;
2. how those hours should be covered by permanent, outsourced, overtime and transferable capacity;
3. which shifts and rest patterns should be rostered; and
4. when recruitment or outsourced-capacity bookings must be initiated.

The four supplied documents support the strategic need for forecast-led capacity planning, probabilistic resource estimates, workforce and contractor management, auditable optimisation and human override. They do **not** provide the operational records required to calculate an EMX baseline, train a productivity model, estimate labour costs or prove improvement against the stated targets. The supplied package contains four PDF reports but no structured operational dataset. Consequently, this report provides a technically executable design and prototype specification, but it does not claim actual staffing accuracy, labour savings, on-time-delivery improvement or AED 0.50 savings per shipment. [filecite\turn0file0](#) [filecite\turn0file1](#) [filecite\turn0file2](#) [filecite\turn0file3](#)

This limitation is consistent with the PDF evidence. The EMX corporate research profile states that authoritative EMX-specific figures were not identified for parcel volume, headcount, courier count, fleet size, warehouse capacity or service-level performance. It also warns that publicly described services do not prove actual productivity, first-attempt delivery or on-time delivery performance. The enterprise platform blueprint similarly states that operational volumes and financial baselines are not public and that ROI must be populated using internal data. (*EMX Corporate, Commercial, Operational and Strategic Research Profile*, pp. 3, 20, 35 and 44; *EMX Enterprise Logistics Platform Blueprint*, pp. 1, 10 and 60). [filecite\turn0file1](#) [filecite\turn0file2](#)

The strongest practical design is a hybrid of:

- a **probabilistic forecast-error layer** around EMX's existing shipment forecast;
- a **store-level productivity and workload model**;
- an **availability and shrinkage model**;
- a **mixed-integer or constraint-programming optimisation layer** for allocation, hiring and shifts;
- an **auditable rules layer** for lead times, legal limits, minimum store coverage and approvals; and
- a planner interface that presents recommendations, uncertainty, cost and service risk.

Hierarchical forecasting is appropriate because EMX demand must remain coherent across network, region, store, day and hour. Forecast reconciliation methods are designed to make independently generated

forecasts add up across such structures, while probabilistic reconciliation can preserve uncertainty distributions rather than producing only point forecasts. <sup>1</sup>

The PDFs themselves point in the same direction: capacity should be calculated from demand forecasts plus productivity and absence distributions and returned as scenario bands rather than a single estimate. They also recommend different horizons for same-day labour movement, shift and subcontractor booking, temporary capacity and longer-term network decisions. (*Enterprise Delivery Management and Network Optimisation Platform Blueprint*, pp. 17 and 19). [filecite turns0file0](#)

The recommended planning cadence is therefore:

| Decision layer          | Horizon                        | Refresh                             | Primary output                                                     |
|-------------------------|--------------------------------|-------------------------------------|--------------------------------------------------------------------|
| Intraday balancing      | Current day to 48 hours        | Event-driven or every 15–60 minutes | Transfers, overtime, shift extensions and outsourced substitutions |
| Roster planning         | One to six weeks               | Daily                               | Store, date, shift and hourly coverage                             |
| Outsourced booking      | Five to ten days ahead         | Daily                               | Outsourced courier quantities and booking deadlines                |
| Permanent recruitment   | Forty-five to sixty days ahead | Weekly, with daily alerts           | Hiring quantities and latest initiation dates                      |
| Strategic workforce mix | Rolling three months           | Weekly                              | Permanent-versus-outsourced capacity trajectory                    |

The principal design conclusion is direct: **the first implementation priority is not a sophisticated neural network. It is a reliable, versioned conversion from forecast shipments to effective courier-hours, followed by a constrained decision model that can be backtested against actual manual plans.**

## Evidence and data audit

### Findings from the supplied research

The documents establish EMX as the Courier, Express and Parcels arm of 7X and describe a broad operating scope that includes domestic express, same-day delivery, international services, freight, fulfilment, returns and specialised logistics. These products have materially different workload characteristics, so “shipment count” alone is not an adequate capacity measure. A same-day, time-windowed, secure or temperature-controlled shipment may consume different courier time from a standard next-day parcel. (*EMX Corporate, Commercial, Operational and Strategic Research Profile*, pp. 1 and 10–14; *EMX Enterprise Logistics Platform Blueprint*, pp. 1–8). [filecite turns0file1](#) [filecite turns0file2](#)

The research identifies public booking, tracking, API, fulfilment and proof-of-delivery capabilities, but does not identify EMX’s current route-optimisation, workforce-management, telematics, data-platform or core parcel-system vendors. This means the prototype should use vendor-neutral input and output contracts rather than assume a specific ERP, WMS, TMS or workforce system. (*EMX Corporate, Commercial, Operational*

and Strategic Research Profile, pp. 19–25; EMX Enterprise Logistics Platform Blueprint, pp. 10–14).

filecite:turn0file1 filecite:turn0file2

The strongest data-design guidance in the PDFs includes a persistent shipment identifier, canonical event vocabulary, versioned location records, controlled exception codes, authoritative vehicle and driver data, shift and contractor relationships, calendars, standard costs, audit events and model versions. Daily route and workforce plans are recommended as intraday or overnight processes, with demand forecasts refreshed daily, weekly or on material events. (*Enterprise Delivery Management and Network Optimisation Platform Blueprint*, pp. 23–25). filecite:turn0file0

## Required internal data inventory

The following data is required before any performance result can be considered valid.

| Data object             | Minimum fields and grain                                                                                     | Required use                                                        | Current status                |
|-------------------------|--------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|-------------------------------|
| Forecast vintages       | forecast_created_at, store, service, date, hour or wave, P50 or point forecast, forecast horizon             | Prevent leakage; reproduce what planners knew at each decision date | <b>Unknown / not supplied</b> |
| Actual demand           | Shipment ID, store, promised date/window, induction time, delivery attempt and completion time, service type | Backtesting, workload calculation and service measurement           | <b>Unknown / not supplied</b> |
| Manual staffing plans   | Plan version, planner, store, date, shift, permanent count, outsourced count, overtime and overrides         | True manual baseline                                                | <b>Unknown / not supplied</b> |
| Rosters and assignments | Courier, employment type, home store, assigned store, start/end, breaks, role and skills                     | Available and scheduled capacity                                    | <b>Unknown / not supplied</b> |
| Attendance and leave    | Planned leave, absence, lateness, training, sickness and actual productive time                              | Shrinkage and availability model                                    | <b>Unknown / not supplied</b> |
| Route execution         | Route, stops, distance, drive time, service time, departure delay, failed attempts and pickups               | Store-specific productivity model                                   | <b>Unknown / not supplied</b> |
| Store master            | Coordinates, operating hours, coverage area, service mix, weekend calendar and minimum opening roster        | Store constraints and geographic transfer feasibility               | <b>Unknown / not supplied</b> |
| Recruitment pipeline    | Request date, approval, offer, permit, joining date, rejection and attrition                                 | Empirical permanent-hire lead-time distribution                     | <b>Unknown / not supplied</b> |

| Data object          | Minimum fields and grain                                                                       | Required use                                 | Current status                |
|----------------------|------------------------------------------------------------------------------------------------|----------------------------------------------|-------------------------------|
| Outsourcing capacity | Supplier, store/zone, requested date, confirmed quantity, no-show rate, price and lead time    | Booking and supplier-allocation optimisation | <b>Unknown / not supplied</b> |
| Labour costs         | Permanent paid-hour cost, overtime, agency rate, recruitment cost, transfer cost and penalties | Cost objective and AED 0.50 test             | <b>Unknown / not supplied</b> |
| Operational KPIs     | On-time delivery, attempt outcome, closure hours, complaints, cancellations and backlog        | Service constraints and benefit validation   | <b>Unknown / not supplied</b> |

The forecast table must preserve **every historical forecast vintage**. Keeping only the latest revised forecast would create leakage because a backtest could use information unavailable to the planner at the historical decision date. Rolling-origin time-series validation requires each training set to contain only observations available before its test period. <sup>2</sup>

If EMX's three-month forecast is daily rather than hourly, the engine must not pretend that hourly forecasts already exist. It should derive hourly demand through a separate intraday-profile model using store, service, weekday, holiday, campaign and seasonality features. That derived layer must be evaluated separately from the original forecast.

## Data-quality tests

The ingestion pipeline should reject or quarantine records that fail the following checks:

| Test               | Failure condition                                                                                | Operational consequence               |
|--------------------|--------------------------------------------------------------------------------------------------|---------------------------------------|
| Key uniqueness     | Duplicate forecast vintage/store/date/hour or duplicate shipment events                          | Double-counted demand                 |
| Temporal integrity | Completion before induction, roster after actual execution, forecast created after target period | Invalid or leaked records             |
| Grain consistency  | Daily forecasts joined directly to hourly staffing without allocation logic                      | False precision                       |
| Missingness        | Missing store, service, shift, courier type, route times or cost rate                            | Unusable capacity or cost calculation |
| Coverage           | Stores or hours with forecasts but no operating calendar or productivity estimate                | Unsupported recommendation            |
| Outliers           | Impossible shift durations, productivity, distances, absence or parcel counts                    | Distorted model estimates             |

| Test                  | Failure condition                                                                            | Operational consequence            |
|-----------------------|----------------------------------------------------------------------------------------------|------------------------------------|
| Status reconciliation | Shipment marked delivered without completion event or courier assigned to overlapping shifts | Invalid capacity and KPI totals    |
| Cost completeness     | Agency or overtime hours without rates                                                       | Understated business case          |
| Versioning            | Overwritten forecasts, rosters or overrides                                                  | Inability to reconstruct decisions |

The minimum useful history depends on the modelling layer. Basic deterministic capacity calculations can start with current forecast, availability and productivity standards, but a reliable demand or productivity model should normally include representative seasonal periods. The supplied blueprint recommends at least 12 months and preferably two to three years for demand forecasting, including major peaks, while allowing simpler rules when history is insufficient. (*Enterprise Delivery Management and Network Optimisation Platform Blueprint*, pp. 24–25). [filecite turn0file0](#)

## Evidence classifications

The following distinctions should be visible in every output:

| Classification | Meaning in this project                                                                         |
|----------------|-------------------------------------------------------------------------------------------------|
| Verified       | Directly observed in supplied internal data, contractual rules or authoritative source material |
| Derived        | Calculated from verified inputs with a disclosed equation                                       |
| Inferred       | Supported by evidence but not explicitly confirmed                                              |
| Assumed        | Configurable placeholder required to run the prototype                                          |
| Unknown        | Material input unavailable and not safely inferable                                             |

The 60/40 mix and the 45–60-day and 5–10-day lead times are **challenge-provided constraints**, not outcomes established by the attached operational data. Their exact interpretation and actual empirical distributions remain to be validated.

## Planning model, baseline and decision logic

### Units and planning grain

The primary demand unit should be **workload-adjusted shipments or stops**, while the primary capacity unit should be **productive courier-hours**.

Raw courier headcount is not sufficient because two couriers can provide different capacity when shift lengths, leave, breaks, attendance, route conditions and store productivity differ. Headcount should be calculated only after required courier-hours have been estimated.

The canonical planning grain should be:

store  $\times$  date  $\times$  hour or operating wave  $\times$  service class

Shift and daily outputs are aggregations of this base grain. Where volume is low, adjacent hours may be grouped into operational waves to reduce unstable ratios and infeasible one-hour shifts.

### Capacity-conversion equations

For store  $s$ , date  $d$ , interval  $h$ , service class  $c$ , and forecast quantile or scenario  $q$ :

$F_{s,d,h,c}^q$  = forecast shipments or stops

$M_{s,d,h,c}$  = workload multiplier for service mix, reattempts, pickups and handling

$$W_{s,d,h}^q = \sum_c F_{s,d,h,c}^q M_{s,d,h,c}$$

Let:

$P_{s,d,h}$  = predicted workload units per productive courier-hour

Then:

$$R_{s,d,h}^q = \frac{W_{s,d,h}^q}{P_{s,d,h}}$$

where  $R_{s,d,h}^q$  is the productive courier-hours required.

Let:

$A_{s,d,h}$  = expected attendance fraction

$B_{s,d,h}$  = productive fraction after breaks, training and non-delivery activity

$U_{s,d,h}^{\max}$  = maximum planned utilisation

Then required scheduled hours are:

$$H_{s,d,h}^q = \frac{R_{s,d,h}^q}{A_{s,d,h} B_{s,d,h} U_{s,d,h}^{\max}}$$

This formulation must be calibrated carefully. If the productivity variable already reflects break time or absence, those effects must not be deducted again. The model dictionary should therefore define productivity explicitly as either:

- shipments per **productive** hour;
- shipments per **paid** hour; or
- shipments per **route operating** hour.

Mixing these definitions would materially bias staffing estimates.

Available effective capacity is:

$$C_{s,d,h} = \sum_e X_{e,s,d,h} \times \Delta_h \times A_{e,d} \times B_{e,d,h} \times U_{s,d,h}^{\max}$$

where  $X_{e,s,d,h} = 1$  if courier  $e$  is assigned to that interval and  $\Delta_h$  is interval duration.

The capacity gap is:

$$G_{s,d,h} = R_{s,d,h}^q - C_{s,d,h}$$

A positive gap indicates expected shortage; a negative gap indicates expected surplus.

## Productivity model

The model should predict expected minutes per shipment or workload units per productive hour using:

- store and zone;
- weekday, weekend and hour;
- route density;
- average route and stop distance;
- building and address complexity;
- service-window mix;
- pickups and returns;
- failed-attempt and reattempt rates;
- parcel size or special handling;
- departure delay;
- traffic and seasonal conditions;
- courier familiarity or tenure, where lawful and operationally justified.

The attached blueprints explicitly caution against judging drivers using raw stops per hour without contextual adjustment. They recommend controlling for density, access complexity, traffic, address confidence, workload feasibility, route familiarity and depot delays. (*Enterprise Delivery Management and Network Optimisation Platform Blueprint*, p. 22; *Delivery Platform Optimization Blueprint*, p. 13).

filecite turn0file0 filecite turn0file3

A pooled hierarchical model is preferable to an independent model for every store. Stores with extensive history can learn local effects, while new or low-volume stores can inherit regional and network priors. A gradient-boosted model with store encodings is practical for an MVP; a hierarchical Bayesian or mixed-effects model offers more explicit pooling and uncertainty but requires greater statistical maturity.

## Forecast uncertainty

A point forecast should not be converted directly into one exact staffing figure. Prediction intervals make forecast uncertainty visible, and the uncertainty generally increases with forecast horizon. <sup>3</sup>

The engine should produce at least:

- **Base:** median or P50 demand;
- **Upper:** operational risk quantile, such as P80 or P90;
- **Stress:** defined disruption scenario.

Permanent capacity should primarily cover stable base demand. Outsourced capacity, overtime and cross-store transfers should cover uncertainty and shorter peaks unless the gap is sustained long enough to justify recruitment.

## Manual baseline

The preferred baseline is the actual historical manual plan:

Manual baseline = planner-approved permanent, outsourced and overtime hours as known at each plan date

That baseline must preserve plan versions and timestamps. Comparing the model against a final adjusted roster would be unfair if the model is evaluated at a 60-day hiring horizon.

If historical manual plans are unavailable, use a labelled **reconstructed baseline**:

$$\hat{R}_{s,d,h}^{baseline} = \frac{F_{s,d,h}}{\text{MedianProductivity}_{network, same weekday, trailing period}}$$

with:

- one network-level productivity factor;
- one historic average absence factor;
- no probabilistic buffer;
- no store-specific productivity model;
- no integrated hiring optimisation; and
- no transfer optimisation.

This is intentionally simple and reproducible. It should not be described as EMX's actual manual method unless confirmed by planner evidence.

## Permanent, outsourced and hiring rules

The 60/40 mix is ambiguous unless its denominator and time aggregation are defined. The recommended primary interpretation is:

$$\text{Permanent mix}_m = \frac{\text{permanent paid courier-hours in month } m}{\text{permanent} + \text{outsourced paid courier-hours in month } m}$$

Paid hours are preferable to headcount because they map directly to capacity and cost and avoid treating unequal shift lengths as equivalent. Until EMX confirms the intended definition, the prototype should also calculate the ratio by headcount and productive capacity.



The 60/40 ratio should usually be a **soft network-month constraint**, not a hard store-hour rule. A hard 60/40 requirement in every store and interval could force unnecessary outsourcing when permanent couriers are available or force excess permanent staffing during temporary peaks.

Permanent recruitment should be recommended only when all of the following are true:

1. the projected shortage begins after the permanent recruitment lead time;
2. the gap persists across a configurable minimum number of weeks;
3. the permanent hire is expected to meet a minimum utilisation threshold after joining;
4. the risk-adjusted permanent cost is lower than the avoided outsourced, overtime and shortage cost over the evaluation horizon; and
5. recruitment does not create an unacceptable long-term surplus under the lower-demand scenario.

A simple persistence rule is:

$$\text{PersistentGap}_{s,w} = \max(0, Q_{\alpha}(R_{s,w}) - C_{s,w}^{\text{existing}})$$

Permanent hires:

$$\text{Hire}_s = \left\lceil \frac{\text{SustainedHoursGap}_s}{\text{contracted weekly hours} \times B \times U^{\max}} \right\rceil$$

Latest hiring initiation date:

$$D_s^{\text{hire}} = D_s^{\text{first persistent shortage}} - L^{\text{perm}}$$

where  $L^{\text{perm}}$  should initially use the conservative end of the challenge range, 60 days, until an empirical lead-time distribution is available.

Outsourced booking date:

$$D_s^{\text{out}} = D_s^{\text{first temporary shortage}} - L^{\text{out}}$$

where the initial conservative booking lead time is 10 days.

Temporary spikes inside the permanent lead-time window should be covered in the following order, subject to feasibility and cost:

1. surplus permanent capacity at the same store;
2. feasible cross-store transfer;
3. approved overtime or flexible shift;
4. already-contracted outsourced capacity;
5. new outsourced booking;
6. controlled unmet-demand slack with a severe penalty and escalation.

Store closure prevention should be a hard constraint:

$$\text{Permanent}_{s,d,h} + \text{Outsourced}_{s,d,h} + \text{TransferIn}_{s,d,h} \geq \text{MinimumOpeningRoster}_{s,d,h}$$

This does not guarantee that all deliveries will be completed, but it prevents the optimiser from reducing a store below the operationally approved minimum solely to save cost.

## Recommended explanation structure

Each recommendation should be generated from structured model facts:

**Store S14, week beginning 18 September:** P80 workload requires 312 productive courier-hours. Existing permanent and confirmed outsourced capacity provide 266 effective hours after planned leave and expected absence, leaving a 46-hour gap. The gap persists for six forecast weeks. Book 34 outsourced hours by 8 September and initiate recruitment for one permanent courier by 20 July. Main drivers: weekend volume uplift, below-network store productivity and two planned leave periods. Confidence is medium because route-time coverage is incomplete.

The explanation should cite actual forecast, productivity, leave, capacity and lead-time values. A language model may render the wording, but it should not invent the rationale or determine the recommendation.

## AI and optimisation architecture

### Recommended model stack

| Layer                    | Recommended method                                                                            | Decision produced                                               | Why it is appropriate                                                                 |
|--------------------------|-----------------------------------------------------------------------------------------------|-----------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Forecast uncertainty     | Residual model by store, weekday, hour and horizon; hierarchical probabilistic reconciliation | P50, upper and stress demand                                    | Uses EMX's existing forecast while quantifying error and preserving network coherence |
| Hourly disaggregation    | Historical intraday profiles or gradient boosting                                             | Store-hour demand from daily forecasts                          | Avoids assuming unsupported hourly granularity                                        |
| Productivity             | Gradient boosting, mixed-effects or hierarchical Bayesian model                               | Expected workload per productive hour and interval              | Captures non-linear and store-specific effects                                        |
| Absence and availability | Empirical distributions initially; calibrated classification when sufficient data exists      | Expected attendance and risk bands                              | Prevents paper headcount from being treated as productive capacity                    |
| Strategic allocation     | MILP or CP-SAT with scenarios                                                                 | Permanent, outsourced, transfer, overtime and hiring quantities | Handles discrete hires, lead times, capacity and mix constraints                      |

| Layer             | Recommended method                               | Decision produced                         | Why it is appropriate                                               |
|-------------------|--------------------------------------------------|-------------------------------------------|---------------------------------------------------------------------|
| Shift generation  | Constraint programming                           | Shift templates and required shift counts | Scheduling involves a very large feasible set and many hard rules   |
| Individual roster | CP-SAT or commercial workforce solver            | Courier-to-shift assignments              | Supports qualifications, rest, preferences and coverage constraints |
| Explanations      | Deterministic reason-code and contribution layer | Recommendation rationale                  | Auditable and resistant to fabricated reasoning                     |

Google's official optimisation documentation identifies constraint programming as well suited to employee scheduling and provides workforce models for shift generation and assignment against coverage requirements and constraints. OR-Tools also supports mixed-integer, constraint, vehicle-routing and graph optimisation. <sup>4</sup>

Route optimisation should remain outside the MVP's main decision scope. Route-derived features such as distance, density and service time should feed the productivity model, but the workforce engine should not turn into a full vehicle-routing platform. Time windows, route duration and shift start/end times are nevertheless relevant inputs because they constrain feasible courier capacity. <sup>5</sup>

## Optimisation formulation

Recommended decision variables include:

$x_{s,d,h}^{perm}$  = permanent courier-hours assigned

$x_{s,d,h}^{out}$  = outsourced courier-hours assigned

$x_{s,d,h}^{ot}$  = overtime hours

$x_{i,j,d,h}^{transfer}$  = hours transferred from store  $i$  to  $j$

$y_{s,t}^{hire}$  = integer permanent hires initiated at time  $t$

$u_{s,d,h}$  = unmet productive-hour requirement

$o_{s,d,h}$  = excess effective capacity

The objective is:

$$\begin{aligned}
\min Z = & \sum c^{perm} x^{perm} + \sum c^{out} x^{out} + \sum c^{ot} x^{ot} + \sum c^{transfer} x^{transfer} \\
& + \sum c^{hire} y^{hire} + \sum \pi^{short} u + \sum \pi^{over} o + \sum \pi^{mix} MixDeviation \\
& + \sum \pi^{change} RosterInstability + \sum_{\omega} p_{\omega} ServiceRisk_{\omega}
\end{aligned}$$

The shortage penalty should be substantially higher than the ordinary paid-hour cost and should increase for premium services, high closure risk and periods where late delivery is likely. Legal, safety and physical requirements should be hard constraints rather than low-weight penalties.

Core constraints include:

$$Capacity_{s,d,h,\omega} + u_{s,d,h,\omega} \geq Requirement_{s,d,h,\omega}$$

$$x_{s,d,h}^{perm} \leq AvailablePermanentHours_{s,d,h}$$

$$x_{s,d,h}^{out} \leq ConfirmedOrBookableOutsource_{s,d,h}$$

$$\sum_j x_{s,j,d,h}^{transfer} \leq TransferableSurplus_{s,d,h}$$

$$y_{s,t}^{hire} \text{ contributes capacity only when } d \geq t + L^{perm}$$

$$OutsourceBooking_{s,t} \text{ contributes capacity only when } d \geq t + L^{out}$$

$$\left| \frac{\sum x^{perm}}{\sum x^{perm} + \sum x^{out}} - 0.60 \right| \leq \epsilon + MixDeviation$$

Shift coverage is represented by:

$$\sum_j a_{j,h} z_{s,d,j} \geq RequiredHeadcount_{s,d,h}$$

where  $z_{s,d,j}$  is the number of couriers assigned to shift template  $j$  and  $a_{j,h}$  indicates whether that shift covers interval  $h$ .

## Handling unseen stores and limited history

For a new store:

1. classify it by geography, service mix, expected density and operating hours;
2. initialise productivity from the closest store cluster or hierarchical prior;
3. widen forecast and productivity intervals;
4. use a higher risk buffer;
5. shorten the recalibration cycle after launch; and
6. require planner approval until sufficient execution history exists.

A new store should not inherit the simple network average without an uncertainty adjustment.

## AI boundaries

Machine learning should predict uncertain variables such as demand error, productivity, absence and service risk. Mathematical optimisation should allocate resources. Deterministic rules should enforce lead times, legal constraints, store minimums and approval policies.

Generative AI should not independently decide:

- how many couriers to hire;
- whether to violate the 60/40 mix;
- whether a store may operate below minimum coverage;
- which worker should be assigned in breach of rest or qualification rules; or
- whether a financial target has been achieved.

NIST's AI Risk Management Framework supports managing AI risk across design, development, deployment and evaluation, with trustworthiness and governance considered throughout the lifecycle. <sup>6</sup>

## Prototype and planner workflow

### Data model

A minimum prototype requires the following tables.

| Table                    | Primary key                                     | Important fields                                                                |
|--------------------------|-------------------------------------------------|---------------------------------------------------------------------------------|
| forecast_vintage         | forecast ID, store, target interval, service    | created timestamp, forecast horizon, P50, P80/P90, source model                 |
| shipment_actual          | shipment ID                                     | store, service, forecastable date, promised window, actual attempts, completion |
| store_interval           | store, timestamp                                | operating flag, minimum roster, holiday, campaign and weather indicators        |
| productivity_observation | store, route or courier, date                   | shipments, stops, productive hours, distance, service time and route mix        |
| courier_master           | courier ID                                      | employment type, home store, skills and contract-hour rules                     |
| roster                   | courier, interval, plan version                 | assigned store, shift, breaks, planned and actual attendance                    |
| leave_absence            | courier, date, event                            | leave type, planned flag, duration and approval                                 |
| outsource_offer          | supplier, store, date, interval                 | quantity, cost, lead time, confirmation and no-show                             |
| recruitment_pipeline     | requisition ID                                  | role, store, initiation, approval, joining and cancellation dates               |
| cost_rate                | workforce type, effective date, store or region | paid-hour, overtime, agency, recruitment and transfer costs                     |

| Table                       | Primary key                | Important fields                                                           |
|-----------------------------|----------------------------|----------------------------------------------------------------------------|
| <code>recommendation</code> | run ID, store, interval    | required hours, allocation, hire quantity, deadlines, risk and explanation |
| <code>override</code>       | recommendation ID, version | planner action, reason, approver, timestamp and outcome                    |

Every model input and output should carry:

- source timestamp;
- effective timestamp;
- data-quality status;
- model or rule version;
- scenario or quantile;
- run ID; and
- evidence classification.

## Core prototype logic

The following scaffold demonstrates the capacity calculation. It is executable once the required columns are populated, but it has not been run against EMX data because no operational dataset was supplied.

```
from __future__ import annotations

from dataclasses import dataclass
from datetime import timedelta
import numpy as np
import pandas as pd

@dataclass(frozen=True)
class PlanningConfig:
    permanent_lead_days: int = 60
    outsource_lead_days: int = 10
    target_utilisation: float = 0.85
    permanent_mix_target: float = 0.60
    persistent_gap_weeks: int = 4
    interval_hours: float = 1.0

REQUIRED_COLUMNS = {
    "store_id",
    "interval_start",
    "forecast_shipments",
    "workload_multiplier",
```

```

        "productivity_per_productive_hour",
        "attendance_fraction",
        "productive_fraction",
        "available_permanent_effective_hours",
        "available_outsourced_effective_hours",
    }

def validate_inputs(df: pd.DataFrame) -> None:
    missing = REQUIRED_COLUMNS.difference(df.columns)
    if missing:
        raise ValueError(f"Missing required columns: {sorted(missing)}")

    bounded = {
        "attendance_fraction": (0.0, 1.0),
        "productive_fraction": (0.0, 1.0),
    }
    for column, (low, high) in bounded.items():
        invalid = ~df[column].between(low, high)
        if invalid.any():
            raise ValueError(f"{column} contains values outside [{low}, {high}]")

    positive = [
        "productivity_per_productive_hour",
        "workload_multiplier",
    ]
    for column in positive:
        if (df[column] <= 0).any():
            raise ValueError(f"{column} must be greater than zero")

def build_capacity_plan(
    inputs: pd.DataFrame,
    config: PlanningConfig = PlanningConfig(),
) -> pd.DataFrame:
    validate_inputs(inputs)
    plan = inputs.copy()

    plan["interval_start"] = pd.to_datetime(plan["interval_start"], utc=True)

    plan["forecast_workload"] = (
        plan["forecast_shipments"] * plan["workload_multiplier"]
    )

    plan["required_productive_hours"] = (
        plan["forecast_workload"]
        / plan["productivity_per_productive_hour"]
    )

```

```

)

effective_fraction = (
    plan["attendance_fraction"]
    * plan["productive_fraction"]
    * config.target_utilisation
)

if (effective_fraction <= 0).any():
    raise ValueError("Effective productive fraction must be positive")

plan["required_scheduled_hours"] = (
    plan["required_productive_hours"] / effective_fraction
)

plan["required_headcount"] = np.ceil(
    plan["required_scheduled_hours"] / config.interval_hours
).astype(int)

plan["existing_effective_capacity"] = (
    plan["available_permanent_effective_hours"]
    + plan["available_outsourced_effective_hours"]
)

plan["capacity_gap_hours"] = (
    plan["required_productive_hours"]
    - plan["existing_effective_capacity"]
)

plan["recommended_additional_hours"] = (
    plan["capacity_gap_hours"].clip(lower=0)
)

plan["surplus_hours"] = (
    -plan["capacity_gap_hours"].clip(upper=0)
)

plan["latest_outsource_booking_date"] = (
    plan["interval_start"]
    - pd.to_timedelta(config.outsource_lead_days, unit="D")
)

plan["latest_permanent_hiring_date"] = (
    plan["interval_start"]
    - pd.to_timedelta(config.permanent_lead_days, unit="D")
)

plan["recommendation_status"] = np.select(

```



```

    [
        plan["capacity_gap_hours"] > 0,
        plan["surplus_hours"] > 0,
    ],
    [
        "SHORTAGE_REQUIRES_DECISION",
        "SURPLUS_REVIEW_TRANSFER",
    ],
    default="BALANCED",
)

return plan

```

The next layer should pass the resulting requirements, availability, costs and lead-time data to a CP-SAT or MILP optimiser. Shift scheduling should be separated from strategic capacity because it contains employee-level rest, qualification and preference constraints, while the strategic model contains hiring, supplier and scenario decisions.

## Planner workflow

The operational workflow should be:

1. ingest the newest forecast vintage and changed workforce information;
2. validate data and identify unsupported stores or intervals;
3. generate base, upper and stress workload scenarios;
4. estimate store-specific productivity and availability;
5. calculate required productive and scheduled hours;
6. optimise permanent, outsourced, overtime and transfer allocation;
7. calculate recruitment and booking deadlines;
8. compare with the previous approved plan;
9. publish only material changes;
10. require planner approval for hiring, major transfers or policy exceptions;
11. record overrides and reason codes; and
12. monitor realised outcomes.

The planner interface should present one network view and one store drill-down.

| View                 | Required content                                                           |
|----------------------|----------------------------------------------------------------------------|
| Network outlook      | Three-month permanent, outsourced and shortage trajectory; mix; cost; risk |
| Store heatmap        | Hourly shortage and surplus by date and store                              |
| Recruitment pipeline | Requisition, requested join date, latest safe initiation date and risk     |
| Outsource bookings   | Supplier capacity, booking deadline, confirmation and no-show risk         |
| Scenario comparison  | P50, upper, stress and planner-adjusted plans                              |

| View                 | Required content                                                      |
|----------------------|-----------------------------------------------------------------------|
| Recommendation panel | Quantity, timing, cost, service impact, confidence and causal drivers |
| Override panel       | Proposed modification, feasibility validation, reason and approval    |
| Performance panel    | Baseline versus model, by forecast horizon and store segment          |

The interface should not overwhelm planners with every small forecast revision. A materiality rule should suppress recommendations unless the new plan changes headcount, hiring, outsourced booking, closure risk or expected cost beyond configurable thresholds.

## Validation, KPIs and business case

### Backtesting design

The backtest must reproduce the historical decision environment.

For every historical decision date  $t$ :

1. select only the forecast vintage available at  $t$ ;
2. select only workforce, leave, cost and recruitment information known at  $t$ ;
3. train models using records before  $t$ ;
4. generate recommendations for the relevant future horizon;
5. freeze the simulated decision;
6. compare it with actual demand, execution and cost; and
7. compare it with the actual manual plan or reconstructed baseline.

Random train-test splitting is inappropriate because it can mix future and past observations. Rolling-origin evaluation is the correct structure for time-dependent forecasts. <sup>2</sup>

Separate backtests are required for:

- next-day and hourly staffing;
- one-to-six-week roster and outsourced decisions;
- 45–60-day recruitment decisions; and
- three-month workforce-mix decisions.

A model that is strong one day ahead may not be suitable for hiring 60 days ahead.

### KPI definitions

#### Demand-capacity match accuracy

Use productive courier-hours rather than headcount:

$$MatchAccuracy = \max \left( 0, 1 - \frac{\sum_{s,d,h} w_{s,d,h} |C_{s,d,h} - R_{s,d,h}|}{\sum_{s,d,h} w_{s,d,h} R_{s,d,h}} \right)$$

This is a weighted accuracy equivalent to one minus weighted absolute percentage error. Report it with:

$$CellCoverage_{\tau} = \frac{\#\{(s, d, h) : |C - R|/R \leq \tau\}}{\#\{(s, d, h) : R > R_{minimum}\}}$$

Low-volume cells should be handled with an absolute tolerance because percentage errors become unstable when required capacity approaches zero.

#### Overstaffed store-hours

$$OverstaffedHours = \sum_{s,d,h} 1 [C_{s,d,h} > R_{s,d,h} + Tolerance_{s,d,h}]$$

#### Understaffed store-hours

$$UnderstaffedHours = \sum_{s,d,h} 1 [C_{s,d,h} < R_{s,d,h} - Tolerance_{s,d,h}]$$

#### Reduction versus baseline

$$Reduction = \frac{BaselineHours - ModelHours}{BaselineHours}$$

This should be calculated separately for overstaffing and understaffing. Combining them can hide a model that reduces overstaffing by accepting serious shortages.

#### Courier utilisation

$$CourierUtilisation = \frac{ActualProductiveCourierHours}{EffectiveAvailableCourierHours}$$

This should not be defined as shipments divided by paid hours unless service mix and route complexity are also reported.

#### Labour cost per shipment

$$LabourCostPerShipment = \frac{PermanentPaidCost + OvertimeCost + OutsourcedCost + TransferCost + RecruitCost}{CompletedEligibleShipments}$$

The AED 0.50 target is achieved only if:

$$BaselineLabourCostPerShipment - ModelLabourCostPerShipment \geq AED\ 0.50$$

while all service, utilisation and closure gates remain satisfied.

The PDF KPI framework similarly defines cost per shipment as attributable operating cost divided by completed shipments and warns against arbitrary cost allocation. It defines on-time delivery using eligible completed shipments and highlights manipulation risks such as excluding delays without governed rules. (EMX Enterprise Logistics Platform Blueprint, pp. 38–40). [filecite turns 0 file 2](#)

### On-time delivery

$$OTD = \frac{EligibleShipmentsCompletedWithinCommitment}{EligibleCompletedShipments}$$

Customer-caused exclusions must use governed reason codes and must not be applied retrospectively to improve the result.

### Store closure

$$StoreClosures = \sum_{s,d,h} 1[ActualOpenStatus = 0 \wedge ClosureReason = WorkforceShortage]$$

The “zero closure” target should count closures caused by workforce shortage. Closures caused by unrelated facility, safety or regulatory events should be disclosed separately.

### Emergency recruitment

Recommended definition:

$$EmergencyRecruitmentRate = \frac{RecruitmentOrAgencyRequestsInitiatedInsideRequiredLeadTime}{AllRecruitmentOrAgencyRequests}$$

The engine should reduce this rate without simply submitting excessive early requisitions.

### Hiring recommendation performance

Use three complementary measures:

- joining-date coverage: whether the recommended hire joined before the projected shortage;
- utilisation after joining: whether the hire achieved acceptable productive utilisation over a defined period; and
- decision regret: the cost difference between the recommended decision and a perfect-hindsight allocation.

### Planner override rate

$$OverrideRate = \frac{MaterialRecommendationsChangedByPlanner}{MaterialRecommendationsPresented}$$

A high override rate may indicate model misalignment; a near-zero rate may indicate poor recording or planner disengagement. Override reasons are more valuable than the percentage alone.

## Statistical validation

Results should be reported by:

- store;
- region;
- hour and operating wave;
- weekday versus weekend;
- forecast horizon;
- permanent and outsourced capacity;
- high- and low-productivity stores;
- high-volume peaks versus normal periods;
- service class; and
- new or low-history stores.

Use block bootstrap or repeated historical periods to estimate uncertainty around improvement, preserving temporal dependence. Report both average performance and worst-decile stores. A network average can hide severe local failure.

## Scenario and sensitivity tests

No numerical scenario output can be calculated without operational inputs. The prototype should nevertheless run the following controlled tests:

| Scenario                        | Expected decision behaviour                                                                                                |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| Demand +10%, +20%, +30%         | Increase P50 and upper capacity; use outsourced capacity for short peaks; trigger permanent hiring only for sustained gaps |
| Wider forecast error            | Increase flexible buffer and reduce irreversible hiring confidence                                                         |
| Weekend spike                   | Adjust store-hour productivity and shift coverage rather than applying a flat network uplift                               |
| Permanent hiring delayed        | Advance recruitment trigger, extend outsourced cover and increase recruitment-risk alert                                   |
| Outsourced capacity constrained | Prioritise transfers and overtime; expose residual service risk rather than assume unlimited supply                        |
| Absence increase                | Raise scheduled hours and reserve capacity; identify stores sensitive to single-person absence                             |
| Productivity decline            | Separate structural route/store problem from ordinary demand growth                                                        |
| New store                       | Use hierarchical prior, wider intervals and mandatory review                                                               |
| Campaign or seasonal peak       | Use event uplift and scenario-specific demand rather than retraining only after the event                                  |

| Scenario                  | Expected decision behaviour                                               |
|---------------------------|---------------------------------------------------------------------------|
| Permanent cost increase   | Recalculate economic break-even and mix deviation                         |
| Outsource price increase  | Shift stable demand toward permanent capacity if lead time permits        |
| Mix target 50/50 or 70/30 | Quantify cost, service risk, flexibility and emergency-recruitment impact |

The sensitivity analysis should rank inputs by effect on total cost, shortage hours and hiring quantity. This identifies whether the business case depends mainly on forecast accuracy, productivity assumptions, absence, supplier price or utilisation policy.

## Conditions for a credible success claim

The prototype may claim progress toward the challenge targets only if:

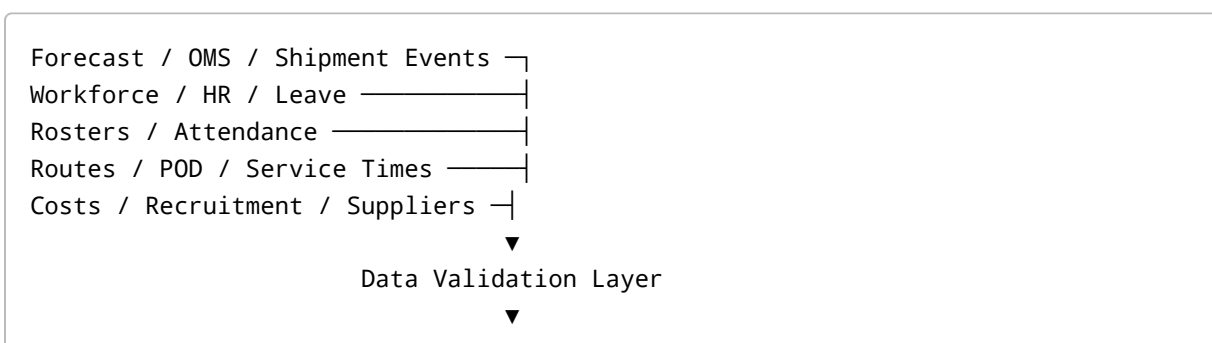
- it is compared with a timestamped manual baseline;
- historical information is separated by decision date;
- staffing requirements are measured in consistent capacity units;
- service levels do not deteriorate;
- costs include permanent, overtime, outsourced, transfer and recruitment components;
- benefits are not double-counted;
- pilot results are evaluated across representative stores and periods; and
- finance validates the AED calculation.

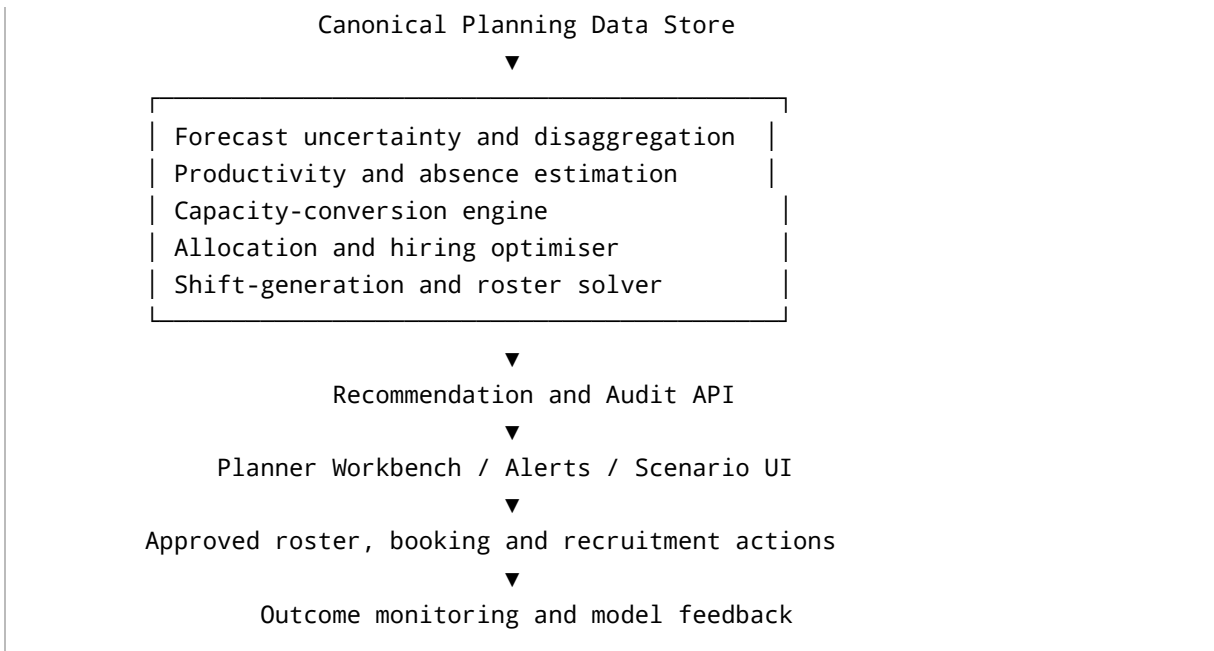
The EMX platform blueprint explicitly states that labour should not be recorded as a cash saving unless headcount, overtime, contractor use or future recruitment is actually reduced. Capacity released for growth should be reported as a capacity benefit, not cash savings. (*EMX Enterprise Logistics Platform Blueprint*, pp. 41–42). [filecite turn0file2](#)

## Production architecture, governance and implementation

### Minimum viable architecture

The production design should remain modular and integration-led.





The attached enterprise blueprint recommends a modular service-oriented architecture with an event backbone, versioned plans, idempotent event processing, active data-quality monitoring and graceful degradation. It also states that existing ERP, WMS, TMS and other systems should remain systems of record where fit for purpose rather than being replaced immediately. (*Enterprise Delivery Management and Network Optimisation Platform Blueprint*, pp. 2 and 26–28). [filecite turn0file0](#)

The capacity-planning prototype does not require a large microservices estate. A practical pilot can use:

- Python services for transformations and models;
- PostgreSQL or an existing warehouse for planning data;
- object storage or a lakehouse for historical snapshots;
- OR-Tools CP-SAT, SCIP, HiGHS or an enterprise solver;
- a lightweight API;
- a planner interface in an approved enterprise framework;
- a model registry;
- scheduled orchestration; and
- immutable recommendation and override logs.

## Security and privacy

Courier identity, attendance, leave, performance and location data require purpose limitation, access controls, retention rules and monitoring. The supplied blueprint specifically identifies UAE personal-data protection requirements, restricted use of driver data and privacy assessment for performance models. (*Enterprise Delivery Management and Network Optimisation Platform Blueprint*, p. 25; *EMX Enterprise Logistics Platform Blueprint*, pp. 35–36 and 59). [filecite turn0file0](#) [filecite turn0file2](#)

The planner should see only the employee-level information needed for rostering. Strategic workforce users normally need aggregated capacity rather than detailed medical or leave reasons.

## Workforce constraints

The production optimiser must be configured against confirmed EMX contracts, policies and applicable legal requirements. UAE official guidance states that normal private-sector working hours are generally eight hours per day or 48 hours per week, with breaks after five consecutive hours and specific overtime provisions; actual applicability and permitted exceptions must be validated by EMX legal and HR teams. <sup>7</sup>

UAE private-sector working hours are reduced during Ramadan, and outdoor work restrictions during specified summer midday periods may affect courier scheduling and effective capacity. These should be represented through an operating calendar rather than manual planner memory. <sup>8</sup>

## Operational ownership

| Responsibility              | Accountable owner            | Required control                                          |
|-----------------------------|------------------------------|-----------------------------------------------------------|
| Forecast approval           | Demand planning              | Forecast vintage, assumptions and campaign sign-off       |
| Store productivity standard | Operations excellence        | Definition, recalibration and exception review            |
| Workforce availability      | HR and store operations      | Roster, leave, absence and employment-type accuracy       |
| Outsourced supply           | Procurement and operations   | Supplier capacity, rates, compliance and no-show data     |
| Recruitment decision        | HR and operations leadership | Requisition approval and joining-date tracking            |
| Cost model                  | Finance                      | Rate definitions, allocation rules and benefit validation |
| Model performance           | Data and product owner       | Drift, coverage, bias and rollback                        |
| Planner overrides           | Planning leadership          | Reason code, approval and post-outcome review             |
| Legal and safety rules      | HR, legal and safety         | Hard constraints and calendar updates                     |
| Data quality                | Named data owners            | Thresholds, remediation and certification                 |

## Implementation path

| Phase             | Recommended deliverable                                         | Validation gate                                                | Main dependency                   | Principal risk              |
|-------------------|-----------------------------------------------------------------|----------------------------------------------------------------|-----------------------------------|-----------------------------|
| Data and baseline | Data contracts, quality report, historical manual-plan baseline | Forecast, roster, attendance, route and cost records reconcile | Access to internal source systems | No timestamped plan history |



| Phase                   | Recommended deliverable                                      | Validation gate                                  | Main dependency              | Principal risk                      |
|-------------------------|--------------------------------------------------------------|--------------------------------------------------|------------------------------|-------------------------------------|
| Capacity model          | Store-hour workload and required-hours model                 | Stable backtest by store and forecast horizon    | Productive-hour definition   | Double-counting shrinkage           |
| Allocation optimisation | Permanent, outsourced, overtime and transfer decisions       | Feasible plans outperform reconstructed baseline | Confirmed costs and policies | Unrealistic shortage penalties      |
| Hiring engine           | Persistent-gap detection and lead-time-aware recommendations | Historical requisition replay                    | Recruitment-pipeline history | Treating all shortages as permanent |
| Planner prototype       | Heatmaps, explanations, scenarios and overrides              | Planner task completion and explanation accuracy | User participation           | Dashboard without decision adoption |
| Shadow pilot            | Recommendations produced but not executed                    | No material safety, coverage or data defects     | Reliable daily refresh       | Planner distrust                    |
| Controlled live pilot   | Approved stores use the engine for selected decisions        | Service, cost and utilisation gates              | Operational change support   | Pilot selection bias                |
| Wider deployment        | Regional rollout and supplier integration                    | Sustained multi-period benefit                   | Integration and ownership    | Scaling before process maturity     |

A realistic pilot should begin with a limited group of stores that includes different demand volumes and productivity profiles but excludes the most operationally fragile locations. Selecting only one unusually stable or high-performing store would not establish network viability.

The pilot should operate in three stages:

- **shadow mode**, where recommendations are compared with planner decisions;
- **assisted mode**, where planners approve or modify recommendations; and
- **controlled automation**, limited to low-risk roster and booking actions within approved thresholds.

Hiring decisions should remain approval-based even after the allocation engine is mature.

## Risks, evidence register and judging assessment

### Principal risks and controls

| Risk                                | Why it matters                             | Required control                                                              |
|-------------------------------------|--------------------------------------------|-------------------------------------------------------------------------------|
| No reliable manual baseline         | Improvement cannot be measured             | Preserve planner plan versions and reconstruct only when necessary            |
| Forecast leakage                    | Produces falsely strong results            | Use forecast vintages and as-of joins                                         |
| Productivity double-counting        | Understates or overstates required hours   | Define paid, route and productive time separately                             |
| Single network productivity average | Hides store differences                    | Store effects, clustering and hierarchical pooling                            |
| Treating headcount as capacity      | Ignores shifts, leave and absence          | Convert everything to effective productive hours                              |
| Hard 60/40 enforcement              | Can increase cost or service risk          | Use network-month soft constraint and report deviations                       |
| Unlimited outsourced assumption     | Creates infeasible plans                   | Supplier-level confirmed capacity and no-show risk                            |
| Hiring for temporary peaks          | Creates long-term surplus                  | Persistence, lead-time and economic break-even tests                          |
| Excessive risk buffer               | Achieves service by overstaffing           | Joint cost and service objective; report overage                              |
| Underpriced shortage penalty        | Produces cheap but unacceptable plans      | Calibrate using service failure and closure consequences                      |
| Generic AI explanations             | Reduces trust and auditability             | Explanations generated only from structured model facts                       |
| Driver productivity gaming          | Encourages unsafe or low-quality behaviour | Context-adjusted productivity and safety/service guardrails                   |
| Planner rejection                   | Good model produces no operational benefit | Shadow mode, overrides and outcome-based feedback                             |
| False savings                       | Capacity benefit reported as cash          | Finance-validated benefit ledger                                              |
| Overly broad platform scope         | Delays the actual workforce use case       | Keep routing, fleet and warehouse modules outside MVP unless needed as inputs |

## Evidence register

| Source                                                                            | Capacity-planning relevance                                                                                                                   | Key pages used                             | Treatment                                                                     |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|-------------------------------------------------------------------------------|
| <i>Enterprise Delivery Management and Network Optimisation Platform Blueprint</i> | Forecast-led capacity planning, productivity and absence distributions, planning horizons, data model, architecture and workforce integration | 1–3, 17, 19, 23–28, 34–35, 38–39 and 47–48 | Strong design evidence; company-specific assumptions must be replaced         |
| <i>EMX Corporate, Commercial, Operational and Strategic Research Profile</i>      | EMX scope, services, public technology evidence and major operational data gaps                                                               | 1–4, 10–25, 35 and 43–45                   | Strong public-source profile; not internal operating evidence                 |
| <i>EMX Enterprise Logistics Platform Blueprint</i>                                | EMX-specific platform architecture, workforce capability, KPI definitions, AI design and lack of public baselines                             | 1–3, 19, 23–25, 30, 38–43 and 53–60        | Strong target-state design; operational ROI remains unknown                   |
| <i>Delivery Platform Optimization Blueprint</i>                                   | Generic forecasting, scheduling, productivity and pilot concepts                                                                              | 1–17                                       | Useful as secondary design material; illustrative figures are not EMX results |

filecite:turn0file0 filecite:turn0file1 filecite:turn0file2 filecite:turn0file3

The fourth document contains generic global benchmarks and illustrative financial assumptions. Its fuel, downtime, redelivery and platform-benefit figures must not be transferred into the EMX business case without independent validation. The document itself labels its financial model illustrative rather than company-specific. (*Delivery Platform Optimization Blueprint*, pp. 13–14). filecite:turn0file3

## Assessment against the judging criteria

| Criterion                  | Current assessment                      | What is required to reach winning evidence                             |
|----------------------------|-----------------------------------------|------------------------------------------------------------------------|
| Depth of AI implementation | <b>Strong proposed architecture</b>     | Train probabilistic error, productivity and absence models on EMX data |
| Problem fit                | <b>Strong</b>                           | Keep scope centred on forecast-to-workforce conversion                 |
| Technical execution        | <b>Design-ready, not data-validated</b> | Implement the data contract, optimiser and historical replay           |
| Measurable improvement     | <b>Unproven</b>                         | Compare with timestamped manual plans using rolling-origin backtests   |

| Criterion                          | Current assessment              | What is required to reach winning evidence                               |
|------------------------------------|---------------------------------|--------------------------------------------------------------------------|
| Explainability                     | <b>Strong proposed approach</b> | Generate explanations directly from model contributions and constraints  |
| Real-world viability               | <b>Moderate to strong</b>       | Confirm source systems, costs, HR rules, suppliers and ownership         |
| Staffing accuracy target           | <b>Not yet measurable</b>       | Define capacity accuracy mathematically and evaluate by store-hour       |
| Twenty-percent imbalance reduction | <b>Not yet measurable</b>       | Establish baseline over- and understaffed store-hours                    |
| Zero closures and OTD above 95%    | <b>Not yet demonstrated</b>     | Add hard minimum roster and validate actual service outcomes             |
| AED 0.50 reduction                 | <b>Not yet demonstrable</b>     | Obtain finance-certified labour costs and completed-shipment denominator |
| Reduced emergency recruitment      | <b>Mechanism defined</b>        | Replay historical requisitions and measure inside-lead-time requests     |

The proposed build is likely to score well only if the prototype demonstrates the full chain:

Forecast vintage → workload → productive hours → effective available capacity → optimised allocation → hiring

A dashboard that stops at forecast visualisation or required headcount will not solve the challenge. A model that predicts required drivers but ignores availability, lead times, supplier capacity, cost and shift feasibility will also fail operationally. The winning prototype should make every recommendation reproducible, expose uncertainty, show its baseline comparison and refuse to claim benefits that the supplied data cannot support.

1 Optimal combination forecasts for hierarchical time series – Rob J Hyndman

[https://robjhyndman.com/publications/hierarchical/?utm\\_source=chatgpt.com](https://robjhyndman.com/publications/hierarchical/?utm_source=chatgpt.com)

2 5.10 Time series cross-validation | Forecasting: Principles and Practice (3rd ed)

[https://otexts.com/fpp3/tscv.html?utm\\_source=chatgpt.com](https://otexts.com/fpp3/tscv.html?utm_source=chatgpt.com)

3 5.5 Distributional forecasts and prediction intervals | Forecasting: Principles and Practice (3rd ed)

[https://otexts.com/fpp3/prediction-intervals.html?utm\\_source=chatgpt.com](https://otexts.com/fpp3/prediction-intervals.html?utm_source=chatgpt.com)

4 REST Resource: scheduling | Operations Research API | Google for Developers

[https://developers.google.com/optimization/service/reference/rest/v1/scheduling?utm\\_source=chatgpt.com](https://developers.google.com/optimization/service/reference/rest/v1/scheduling?utm_source=chatgpt.com)

5 Time windows | Route Optimization API | Google for Developers

[https://developers.google.com/maps/documentation/route-optimization/concepts/time-windows?utm\\_source=chatgpt.com](https://developers.google.com/maps/documentation/route-optimization/concepts/time-windows?utm_source=chatgpt.com)

6 AI Risk Management Framework | NIST

[https://www.nist.gov/itl/ai-risk-management-framework?utm\\_source=chatgpt.com](https://www.nist.gov/itl/ai-risk-management-framework?utm_source=chatgpt.com)

7 8 Working hours and overtime | The Official Platform of the UAE Government

[https://u.ae/en/information-and-services/jobs/employment-in-the-private-sector/working-hours?utm\\_source=chatgpt.com](https://u.ae/en/information-and-services/jobs/employment-in-the-private-sector/working-hours?utm_source=chatgpt.com)